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Revolutionizing patient care: assessing ai-created discharge education for total hip replacement patients

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Abstract

Background Previous research has shown that artificial intelligence (AI) technology is effective in a wide range of applications. Although studies exist in other specialties evaluating AI-generated educational materials, no study was found focusing on Turkish-language discharge education materials specifically for total hip replacement patients. This study was conducted using the ChatGPT 4.0 program for patients who underwent total hip replacement to develop an artificial intelligence-supported discharge education material that patients who will be discharged after surgery can read and understand.

Methods In this methodological study, expert opinions were obtained while preparing the artificial intelligence-assisted discharge education material for patients who underwent AI-assisted discharge education, and the educational needs of the patients were determined by reviewing the literature. Then, 6 commands were given in the ChatGPT-4 program to create the educational content. The Turkish readability of the created educational content and the readability of the health literature were evaluated using formulas. The validity of the total hip replacement discharge material was presented to 10 experts. Experts assessed the comprehensibility and applicability of the educational material using the Patient Education Materials Assessment Tool and the Global Quality Scale, which assesses the quality of educational materials.

Results It was concluded that the readability index of the total hip replacement discharge education material for Turkish was 69.5, and the Turkish readability level was “easily understandable”. The readability index of health literature was 9.30, 7.90, 8.09, 60.0, and 10.00, respectively. When the internal consistency coefficient between the experts was examined, it was 0.84. It was determined that the Patient Education Materials Evaluation Tool understandability score average was 80.60 ± 9.74 , and the actionability score average was 77.4 ± 19.10 . The Global Quality Scale (GQS) score average was found to be 4.20 ± 0.79 (range: 3–5).

Conclusion At the end of the study, it was determined that the educational material was adequate in terms of understandability and actionability. The Turkish readability level was also reasonable and easily understandable.

Keywords Artificial intelligence, ChatGPT, Educational material, Patient education, Total hip replacement

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Introduction

Total hip replacement is a common and effective surgery for patients with severe pain and movement limitations due to degenerative hip joint damage. The primary indication is osteoarthritis, which causes functional loss and reduced quality of life. Osteoarthritis, a degenerative arthritis, results from the wear and tear of cartilage, impairing joint function and causing pain, restricted movement, and difficulty in daily activities such as walking and sleeping. This condition often leads to social isolation due to limited participation in daily life [1–4]. According to the 2023 Health Statistics of the Organisation for Economic Co-operation and Development (OECD), the average total hip replacement (THR) rate is 172 per 100,000 individuals, with an increasing number of surgeries performed annually. The countries with the highest THR rates are Sweden (320/100,000), Germany (300/100,000), and Austria (280/100,000), while the rate in Türkiye is 64/100,000 [2]. Despite its growing prevalence and high overall success rates, total hip replacement (THR) remains a major surgical procedure associated with potential complications such as postoperative pain, reduced mobility, prolonged recovery periods, and delayed return to daily functioning [1, 5]. Recent large-scale studies have indicated that the 90-day mortality rate following THR is approximately 0.3% (3 per 1,000 cases), which is considerably lower than previously reported estimates of 1–2% [6]. Furthermore, while overall patient satisfaction is generally high, approximately 7% of individuals report dissatisfaction with their surgical outcomes [7]. These findings underscore the need for comprehensive, accessible, and patient-centered educational materials to help patients develop realistic expectations and support successful postoperative adaptation. The recovery process begins immediately after surgery and continues for years. However, many patients lack adequate information about post-operative care. Research suggests that preoperative education enhances patient satisfaction and improves outcomes. Informed patients are more likely to engage in their care [8, 9]. Psychological, emotional, and physical preparation is essential in the preoperative period, as insufficient information can cause anxiety and concerns regarding surgery and recovery [4, 10]. Studies show that effective preoperative education can shorten hospital stays, reduce postoperative painkiller use, decrease anxiety, and increase satisfaction [1, 4, 9, 11]. Tailored educational materials that consider the patient's surgery type, literacy level, and cultural background, along with printed and personalized resources, improve patient compliance and satisfaction [9, 12].

With the rapid evolution of scientific literature and technology, experts are increasingly relying on AI to enhance accessibility and convenience [13]. AI programs

like ChatGPT, launched by OpenAI in 2022, generate complex responses based on human prompts, utilizing large data sets and computational techniques. ChatGPT, which has gained global attention, is capable of producing human-like text, offering insights on a wide range of topics, and serving an international audience due to its multilingual capabilities [14, 15]. While ChatGPT is effective for creating content, providing information, and offering explanations, its lack of evidence-based information raises concerns about accuracy. Its widespread use, particularly in the healthcare sector, brings ethical issues and questions regarding data reliability [14, 16]. The increasing reliance on online sources is particularly evident during times of restricted access to healthcare services. For instance, during the COVID-19 pandemic, YouTube emerged as a widely used platform for stroke rehabilitation information. However, the majority of videos were found to be of low quality and contained insufficient data, as reported by Ozduran and Hanci (2023). This highlights the critical importance of ensuring the accuracy, quality, and trustworthiness of health-related information disseminated through digital media platforms [17].

Patient education plays a vital role in the surgical process by reducing anxiety, improving compliance, and increasing awareness of potential complications. To be effective, materials must match the patient's level of understanding [18]. With the growing use of online resources, concerns remain about the accuracy and quality of such content, as misinformation can negatively impact patient outcomes [19].

To promote health equity, major health authorities—including the National Institutes of Health (NIH), the U.S. Department of Health and Human Services (HHS), and the American Medical Association (AMA) recommend that patient education materials be written at or below a 6th-grade reading level. However, many online resources fall short of this standard. For instance, a study analyzing 95 websites on chronic low back pain found that the average readability was classified as “very difficult” (FRES: 55.74; GFOG: 12.76), and reliability was low, with only 28.4% meeting JAMA criteria and 33.7% adhering to HONcode standards [20]. These results highlight the need for accessible, high-quality educational materials. Today, the increasing use of artificial intelligence (AI) offers new opportunities to improve the accessibility and personalization of health education. Meanwhile, YouTube remains one of the most frequently used platforms for obtaining health-related information. However, many videos on this platform lack scientific accuracy and professional oversight. For instance, Kara et al. (2024) evaluated YouTube videos related to inflammatory back pain and found that most were of low quality and lacked reliability. This highlights the urgent need for high-quality,

expert-reviewed educational materials, especially as patients increasingly turn to digital platforms for medical guidance [21]. Similarly, during the COVID-19 pandemic, YouTube became a common source of stroke rehabilitation information, but studies reported that the majority of content was insufficient and of poor quality [17]. These findings highlight the urgent need for accurate, understandable, and professionally validated educational materials—especially as digital platforms become a primary source of health information for the public.

The aim of this study is to assess the comprehensibility and applicability of AI-generated discharge education materials developed in Turkish for patients undergoing total hip replacement surgery. Although studies evaluating AI-generated content exist in other medical specialties and languages, this study aims to contribute to the growing literature by focusing specifically on Turkish-language discharge education materials tailored for patients undergoing total hip replacement surgery. While this may not be the first use of AI for educational content creation in healthcare, its novelty lies in addressing a linguistically and clinically specific gap.

Methods

Design

This study utilized a methodological and cross-sectional design to develop and evaluate artificial intelligence-supported discharge education materials for patients undergoing total hip replacement. The study was conducted between February and March 2025. While preparing artificial intelligence-supported discharge education material for patients who underwent total hip replacement, the educational needs of the patients were first determined. Then, the commands of the ChatGPT 4.0 program are included in the scope of the transferred education content. First, the readability of the created training content was evaluated. Expert opinion was obtained for the final version of the draft.

Table 1 Educational material topics for patients undergoing surgery after total hip replacement

1. I am an orthopedic nurse. Please provide discharge education for a patient who has had total hip replacement surgery.
2. I am an orthopedic nurse. Please provide education on the Post-Operative Recovery Process for patients who have undergone total hip replacement surgery.
3. I am an orthopedic nurse. Please provide education on Pain Management for patients undergoing total hip replacement surgery.
4. I am an orthopedic nurse. Please provide training on Movement and Activities for patients who have had total hip replacement surgery.
5. I am an orthopedic nurse. Please educate patients who have had total hip replacement surgery about what to consider
6. I am an orthopedic nurse. Please educate patients who have total hip replacement surgery about possible complications.

Determining the educational needs of patients

To determine the educational needs of patients, the researchers gathered input from an orthopedic specialist, orthopedic clinic nurses, and academic nurses regarding the specific needs of patients undergoing total hip replacement. Additionally, a literature review was conducted to identify the information required by patients for post-operative care following total hip replacement [11, 22, 23].

Creation of THR discharge training material content in the ChatGPT 4.0 program

By conducting a systematic review of the literature on post-total hip replacement care, the researchers identified seven key sources, including systematic reviews, meta-analyses, and evidence-based postoperative care protocols [1, 5, 9, 22, 24–26]. These studies were selected based on methodological rigor, clinical relevance, and alignment with current orthopedic nursing practices. The selected resources were then synthesized and loaded into the ChatGPT 4.0 program, where prompts were generated to create educational content under the titles specified in Table 1. The resulting booklet consisted of 16 pages and 2207 words. Although six initial prompts were used to generate the educational content, the process was iterative. Each output was reviewed by the researchers for clinical accuracy, completeness, and clarity. If any prompt yielded content that was inconsistent, overly complex, or lacked key postoperative care elements, it was revised and re-entered into ChatGPT with additional clarifications. This iterative prompting approach helped mitigate known variability in LLM outputs and ensured that the final content aligned with patient education standards and expert expectations. All prompts were entered into the ChatGPT 4.0 interface in Turkish, and the responses were also generated in Turkish. This was explicitly ensured by using Turkish-language input and reviewing the initial outputs for language consistency. The use of Turkish throughout the content creation process is essential to ensure that the readability assessments using the Ateşman Index and other Turkish-specific measures are valid and reflective of actual language characteristics [27]. Had the prompts or responses been in English and later translated, the results could have introduced linguistic distortion and undermined the methodological reliability of the readability scores. Therefore, conducting both prompt input and response generation in Turkish eliminates potential bias due to translation and ensures methodological rigor.

To minimize potential risks associated with AI-generated content, such as hallucinations, inaccuracies, and safety issues, we implemented a multi-step validation process. All discharge education materials generated by ChatGPT were reviewed by the research team for factual

accuracy, consistency with current clinical guidelines, and alignment with established orthopedic nursing standards. In addition, two independent experts—a senior orthopedic nurse and an orthopedic surgeon—cross-checked the content against peer-reviewed literature and evidence-based protocols. Any information deemed inaccurate, unsupported by evidence, or potentially misleading was revised or removed prior to inclusion in the final educational materials.

Prompt refinement strategy and iterative workflow

Although six initial prompts (as shown in Table 1) were used to structure the educational content, the generation process was iterative and involved systematic refinements. Each response generated by ChatGPT-4.0 in Turkish was carefully reviewed by two academic surgical nurses for clinical accuracy, completeness, and alignment with national orthopedic care standards. When deficiencies were noted—such as missing critical information, overly technical language, or insufficient detail—follow-up prompts were crafted with added specificity. Example refinements included:

“Revise this text for easier understanding by a layperson.”

“Please include step-by-step instructions for using a walker after hip surgery.”

“Add visual cues (e.g., bullet points) to explain home care tasks.”

“Translate the explanation into more accessible Turkish, avoiding medical terminology where possible.”

This feedback loop continued until the outputs met the standards for readability, clarity, and actionability. No manual rewriting or use of external editing software was

involved; however, the prompt development process was guided by clinical expertise and validated through relevant literature.

Figure 1a. AI-generated image depicting post-operative hip anatomy created by DALL-E.

Figure 1b. AI-generated image illustrating assisted mobility after THR

Evaluation of readability

Readability is a concept that refers to the ability of printed educational materials to be easily understood and followed by readers. Ateşman developed the readability formula for Turkish by adapting the Flesch formula. According to Ateşman, “The average sentence length in Turkish is 9–10 words, and the average word length is 2.6 syllables. According to the formula, the readability level is between 0–100. As the obtained score approaches 100, the readability of the text becomes easier, and as it approaches 0, the readability of the text becomes more difficult [28] (<http://okunabilirlikindeksi.com>). It was evaluated using the “Simple Measure of Gobbledygook (SMOG)”, “Flesch-Kincaid”, and “Flesch Reading Ease” formulas frequently used in the health literature by Wang and colleagues [28]. Finally, it was evaluated using the “Automatic Readability Index” formula developed by Smith and Senter [29] (<https://readabilityformulas.com>).

Readability levels were interpreted in accordance with thresholds reported by Ozduran et al. (2025) [30]. Accordingly, an FRES score of 80 or above is considered acceptable for general understanding, while a score of 6 or below is regarded as the target threshold for the other six formulas (e.g., FKGL, SMOG, ARI). Scores above 6 in these formulas indicate that the material may be difficult for individuals with limited literacy. In our analysis, most scores—except for the Ateşman index—exceeded the

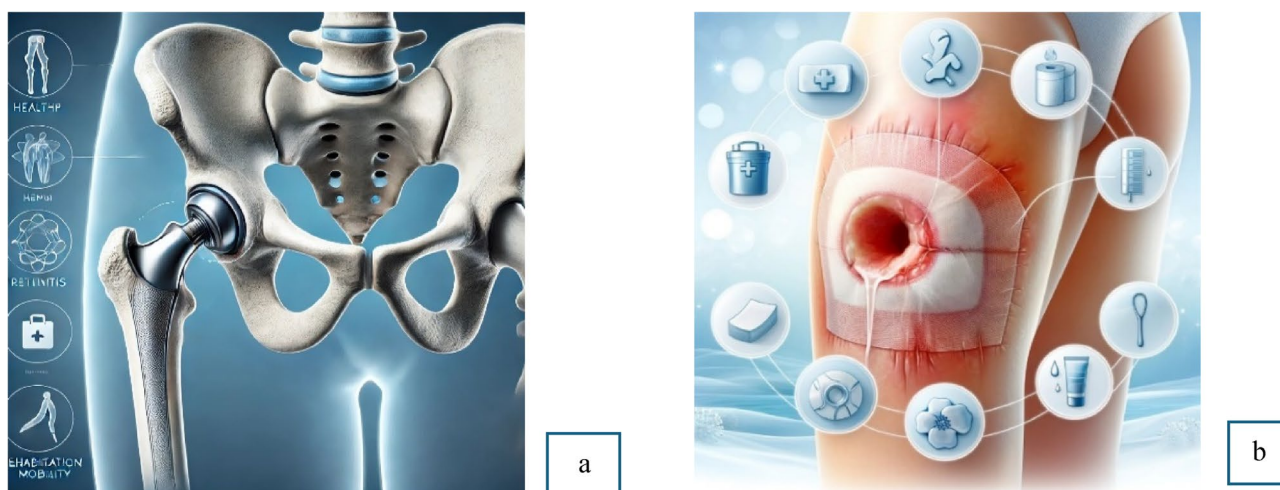


Fig. 1 AI-generated images created by DALL-E: (a) Post-operative hip anatomy; (b) Assisted mobility after total hip replacement

6th-grade level, indicating that the material was moderately to fairly difficult to read.

Obtaining expert opinions on the prepared educational material

The validity of the artificial intelligence-supported discharge education material for patients undergoing total hip replacement was presented to the opinion of 10 experts who have doctoral degrees in surgical nursing and conduct studies in the field of orthopedics. These experts evaluated the educational material’s understandability and actionability using the Patient Education Materials Assessment Tool (PEMAT-P) and its quality using the Global Quality Scale. The number of experts consulted in this study ($n=10$) is consistent with recommendations for content validation studies, where 6 to 10 experts are generally considered sufficient for reliable qualitative evaluation and consensus building. This range aligns with previous studies employing the Patient Education Materials Assessment Tool (PEMAT) and is supported by methodological guidelines for instrument validation in health research [15, 31].

Patient education materials assessment tool (PEMAT)

The Patient Education Materials Evaluation Tool (PEMAT) was developed by Shoemaker and colleagues in 2014 [32] to evaluate and compare the comprehensibility and applicability of printable and audiovisual educational materials. Understandability: Patient education material is considered understandable if individuals with varying levels of education and health literacy can understand and explain the key messages. Actionability: When individuals with different levels of education and health literacy can determine what they can do based on the information provided, the patient education material is applicable. The Turkish validity and reliability of the PEMAT was made by Akkoç and Orgun in

2023 [18]. PEMAT has two versions: Printable Materials Patient Education Materials Evaluation Tool (PEMAT-P) and Audiovisual Materials Patient Education Materials Evaluation Tool (PEMAT-A/V). The printable materials consist of a total of 26 items for PEMAT-P, 17 items evaluating understandability, and 7 items evaluating applicability. The scale is scored as “0” (disagree), “1” (agree), and “Not applicable.” Scoring is obtained by dividing the total score by the possible score and multiplying by 100. The final score is evaluated between 0 and 100 in terms of comprehensibility and applicability. The higher the score, the more understandable or applicable the material is. In this study, the PEMAT-P was used to evaluate the comprehensibility and applicability of artificial intelligence-supported discharge education material in patients who underwent total hip replacement. Cronbach alpha reliability coefficient for PEMAT-P has been reported as 0.901 (18). In our study, Cronbach’s alpha reliability coefficient was found to be 0.76.

Quality score of the education materials (GQS)

The appropriateness and quality of the content of the artificial intelligence-supported discharge education material for patients undergoing total hip replacement were evaluated using the GQS. GQS was developed by Bernard et al. (2007). It is rated using a five-point Likert scale [32]. Scores refer to the quality of the educational material and the extent to which the assessor found it helpful to patients. Accordingly, a score of 1 indicates poor quality and a score of 5 indicates excellent quality (Table 2) [33–35].

Statistical analysis

Statistical analyses were performed using SPSS 25.0 (Statistical Package for Social Science). Descriptive statistics were presented as unit count (n), percentage (%), and mean $\pm$ standard deviation ($\bar{X} \pm SD$). The intraclass correlation coefficient (ICC) was used to assess inter-rater reliability among expert evaluations. Internal consistency of the PEMAT-P scores was evaluated separately using Cronbach’s alpha. The ICC was calculated based on a two-way random-effects model (absolute agreement), while Cronbach’s alpha was computed for all PEMAT items combined, as well as separately for the Understandability and Actionability subscales. $p < 0.05$ was taken as the significance level.

Ethical statement

This study did not require ethics committee approval because it did not involve any implementation within an institution, patient recruitment, or the collection of patient data. Only an open-access AI tool was used and voluntary expert feedback was obtained. Therefore, no

Table 2 GQS criteria GQS Description: the global quality scale (GQS) is a 5-point likert scale used to evaluate the overall quality, logical flow, and usefulness of health education materials. Higher scores indicate better quality and greater usefulness for patients

Criteria	Scores
Poor quality, poor flow of the site, most information missing, not at all useful for patients	1
Generally poor quality and poor flow, some information listed but many important topics missing, of very limited use to patients	2
Moderate quality, suboptimal flow, some important information is adequately discussed but others poorly discussed, somewhat useful for patients	3
Good quality and generally good flow, most of the relevant information is listed, but some topics not covered, useful for patients	4
Excellent quality and excellent flow, very useful for patients	5

ethics committee approval was required according to national research ethics guidelines.

Results

Readability evaluation

It was determined that the Turkish readability index of the developed total hip replacement discharge training material was 69.5 and was ‘easily understandable’. The Automated Readability Index (ARI) was 9.30 (Slightly Difficult), the Simple Measure of Gobbledygook (SMOG) was 7.90 (Average - Slightly Difficult), the Flesch-Kincaid Grade Level was 8.09 (Average - Slightly Difficult), and the Flesch Reading Ease score was 60.0 (Fairly Difficult), all of which are frequently used in health literature. The average reading level consensus was 10.00 (Somewhat Difficult) (Table 3).

Inter-rater reliability

In our study, the internal consistency coefficient between experts was 0.84 (95% CI [0.271–0.927], $p < 0.05$).

Precision of estimates

In several parameters, the 95% confidence intervals (CIs) were relatively wide. This imprecision is likely related to the limited sample size ($n = 10$ experts) and variability in expert scoring, as reflected by the standard deviations in PEMAT and GQS scores. The relatively broad CIs, such as the ICC 95% CI [0.271–0.927], also indicate heterogeneity in expert ratings. While these findings do not invalidate the overall conclusions, they highlight the importance of larger sample sizes and more homogeneous expert panels in future studies to obtain narrower, more precise CIs.

Patient education materials assessment tool (PEMAT)

In most of the PEMAT items (15 items), 90% and above of the experts responded that they agree. The PEMAT understandability score average was determined as 80.60 ± 9.74 (64–100) and the PEMAT applicability score

average was determined as 77.4 ± 19.10 (40–100). The Global Quality Scale score average, which evaluates the appropriateness and quality of the content of the total hip replacement discharge training material, was 4.20 ± 0.79 (3–5) (Table 4, Fig. 2).

Discussion

Total hip replacement (THR) is a common orthopedic surgery, yet it remains associated with notable risks such as complications, delayed recovery, and patient dissatisfaction. Effective patient education is essential to enhance compliance, reduce anxiety, and improve postoperative outcomes [1, 5]. In recent years, the integration of AI into healthcare has created new opportunities for developing scalable, accessible, and patient-centered education tools [36–38].

Studies have shown that adapting existing orthopedic patient education materials to recommended readability levels is both feasible and effective, while still maintaining necessary detail and accuracy [39]. However, most of these studies utilized traditional materials rather than AI-generated content. In contrast, this study developed AI-supported discharge education materials for THR patients, addressing a growing demand for digital health information. Today, patients increasingly rely on internet-based platforms and AI tools to access surgical, procedural, and aftercare information. As such, the readability and clarity of online health content play a critical role in enabling patients to understand and apply this information effectively [40].

In our study, the Turkish readability index of the artificial intelligence-supported discharge training material was calculated as 69.5, which places it near the threshold of being considered “easily understandable” according to the Ateşman Index. However, widely used international metrics such as the Flesch-Kincaid Grade Level (8.09), Automated Readability Index (9.30), and Flesch Reading Ease (60.0) classified the material as difficult. This discrepancy highlights a borderline readability level, suggesting that the content may not be adequately tailored for individuals with low health literacy. Future revisions will aim to align the text with a 6th–8th grade reading level through shorter sentences, simpler language, and improved structure using bullet points, headings, and visual supports. Planned improvements include shortening sentence length, replacing complex medical terms with plain language, and enhancing structural clarity through bullet points, headings, and visual support. These refinements will aim to reduce the Flesch-Kincaid Grade Level, SMOG, and ARI scores, ensuring that the content is not only culturally and linguistically appropriate but also accessible to individuals with varying levels of health literacy.

Table 3 Evaluation of readability

	Readability index	Readability Level
Ateşman	69.5	9th or 10th grade students (Easy to understand)
Automated Readability Index (ARI)	9.30	10th Grade Slightly Difficult
Simple Measure of Gobbledygook (SMOG)	7.90	8th Grade Average - Slightly Difficult
Flesch-Kincaid Grade Level (FKGL)	8.09	8th Grade Average - Slightly Difficult
Flesch Reading Ease (FRE)	60.0	10th to 12th Grade Fairly Difficult
Average Reading Level Consensus (ARLC)	10.0	10th Grade Somewhat Difficult

Table 4 Patient education materials assessment tool (PEMAT) expert opinions

Scale Items	Agree n (%)	Disagree n (%)	Not appli- cable n (%)
Understandability Subscale (19 Items)			
1. The material makes its purpose completely evident	10 (100)		
2. The material does not include information or content that distracts from its purpose	10 (100)		
3. The material uses common, everyday language	10 (100)		
4. Medical terms are used only to familiarize the audience with the terms. When used, medical terms are defined	9 (90.0)	1 (10.0)	
5. The material uses the active voice	10 (100)		
6. Numbers appearing in the material are easy to understand	10 (100)		
7. The material does not expect the user to perform calculations	10 (100)		
8. The material breaks or “chunks” information into short sections	10 (100)		
9. The material's sections have informative headers	10 (100)		
10. The material presents information in a logical sequence	9 (90.0)	1 (10.0)	
11. The material provides a summary	8 (80.0)	1 (10.0)	1 (10.0)
12. The material uses visual cues (e.g., arrows, boxes, bullets, bold, larger font, highlighting) to draw attention to critical points	7 (70.0)	3 (30.0)	
13. The material uses visual aids whenever they can make content more easily understood (e.g., illustration of healthy portion size)	6 (60.0)	4 (40.0)	
14. The material's visual aids reinforce rather than distract from the content	3 (30.0)	7 (70.0)	
15. The material's visual aids have clear titles or captions	2 (20.0)	7 (70.0)	1 (10.0)
16. The material uses illustrations and photographs that are clear and uncluttered	3 (30.0)	7 (70.0)	
17. The material uses simple tables with short and clear row and column headings	2 (20.0)	1 (10.0)	7 (70.0)
18. The material uses illustrations and photographs that are clear and uncluttered	3 (10.0)	7 (70.0)	
19. The material uses simple tables with short and clear row and column headings	2 (20.0)	1 (10.0)	7 (70.0)
Actionability Subscale (7 Items)			
20. The material identifies at least one action the user can take	10 (100)		
21. The material addresses the user directly when describing actions	10 (100)		
22. The material breaks down any action into manageable, explicit steps	8 (80.0)	2 (20.0)	
23. The material provides a tangible tool (e.g., menu planners, checklists) whenever it could help the user take action	7 (70.0)	3 (30.0)	
24. The material provides simple instructions or examples of how to perform calculations	5 (50.0)		5 (50.0)
25. The material explains how to use the charts, graphs, tables, or diagrams to take actions		1 (10.0)	9 (90.0)
26. The material uses visual aids whenever they can make it easier to act on the instructions	2 (20.0)	7 (70.0)	1 (10.0)
	Mean ± SD	Min	Max
PEMAT understandability (0–100)	80.60 ± 9.74	64	100
PEMAT actionability (0–100)	77.4 ± 19.10	40	100
GQS Criteria (1–5)	4.20 ± 0.79	3	5

SD Standard Deviation

“Note: PEMAT scores were calculated as: (Total number of ‘Agree’ responses/Total number of scored items excluding ‘Not Applicable’) × 100. Results are presented as percentages”

Supporting these findings, a recent study by Ozduran et al. (2025) evaluated the readability of AI-generated Turkish content related to low back pain. They reported that the materials had low Ateşman Index scores, indicating they were difficult to read and not well-suited for the general population [30]. Similarly, an earlier study evaluating Turkish internet-based patient education materials on low back pain also found that readability levels exceeded recommended thresholds for public comprehension [41]. These results reinforce our conclusion that AI-generated Turkish materials often fail to meet recommended readability thresholds unless careful prompt

optimization and content tailoring are applied. Considering the shared musculoskeletal focus of both studies, this alignment further underlines the need for prompt engineering and linguistic adjustments to improve patient comprehension [30, 41].

The readability of patient education materials should be tailored to the level of the intended audience. In this study, the readability level was found to be approximately at the 11th–12th grade level, which is considered high but, as reported in similar studies, is within an acceptable range. Rooney et al. (2021) defined a grade level range of 11 to 14 as indicative of the literacy required to

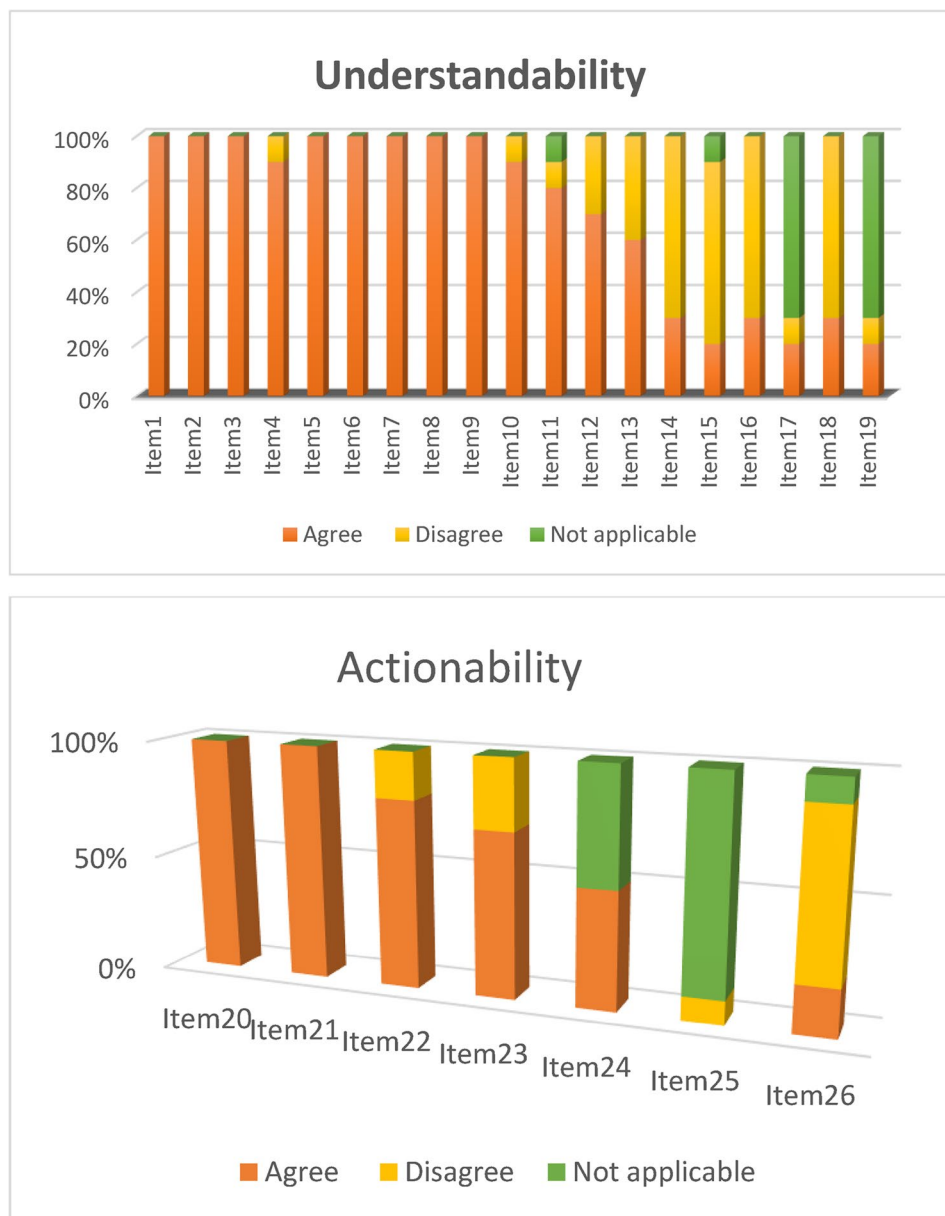


Fig. 2 PEMAT Understandability and Actionability Expert Opinions Graph

comprehend scientific publications, based on their analysis of patient education materials in high impact medical journals [42]. Despite these limitations, the Global Quality Scale average score of 4.20 ± 0.79 (range: 3–5) indicates that the educational material is perceived as high-quality by experts.

In a study using ChatGPT-3.5, the readability of the generated educational material was found to align with the 6th–7th grade level, with favorable scores across FRE (76.9 ± 1.2), FKGL (5.9 ± 0.8), SMOG (8.8 ± 0.6), and GFI (7.7 ± 0.9) indices [43]. Similarly, efforts to reduce the readability level of materials on lumbar disc herniation, scoliosis, and stenosis surgeries through targeted

commands led to significantly improved readability [39]. A separate study using ChatGPT-4.0 initially produced materials with low readability scores, which were substantially improved through refined prompt instructions [44]. These findings highlight the importance of effective prompt engineering in tailoring AI-generated content to meet patient literacy needs. Furthermore, structured and specific prompt usage has been shown to significantly improve both readability and user comprehension in AI-generated educational content, as confirmed by recent empirical studies [35, 42].

Furthermore, although digital platforms such as YouTube are increasingly used for patient education due to

their accessibility, the accuracy and quality of the content they provide are often questionable. For example, Kocyigit et al. (2023) evaluated the reliability and quality of YouTube videos as a source of information for transcutaneous electrical nerve stimulation (TENS)—a treatment method frequently used for hip pain—and found that most videos were of low quality and lacked reliability. This finding underscores the need for professionally reviewed, evidence-based educational materials—especially as patients frequently turn to such platforms for health information [45].

Despite the inclusion of DALL-E-generated visuals in the current study, expert evaluations identified visual aids as the weakest aspect of the material. Elements such as clarity, relevance, and presence of informative captions were rated poorly, suggesting that the visuals did not adequately support the text. This is particularly concerning given that visual aids are crucial for improving comprehension, especially for individuals with limited health literacy. The underperformance may be attributed to the generic design and limited contextual relevance of AI-generated images. Future iterations should incorporate professionally designed visuals developed in collaboration with healthcare or visual communication experts. Research consistently shows that well-aligned, high-quality visuals can significantly enhance patient understanding and information retention [33, 40]. In our study, 10 academic experts evaluated the comprehensibility and applicability of the materials. The mean PEMAT-P comprehensibility score was 80.60 ± 9.74 (range: 64–100), and the applicability score was 77.4 ± 19.10 (range: 40–100). These results show that the training material is well understandable and applicable.

Although the average applicability score was relatively high (77.4), the large standard deviation (± 19.10) and the minimum score (40) indicate substantial variability among expert evaluations. This suggests that certain sections of the material were perceived as less actionable or not directly relevant by some reviewers. Feedback revealed that these lower scores were primarily due to limitations in visual aids and the absence of interactive or step-by-step tools, rather than issues with content accuracy.

Implications of wide confidence intervals

The relatively wide confidence intervals observed in some of our results, particularly the ICC (95% CI [0.271–0.927]), indicate a degree of imprecision in the estimates. This is likely due to the limited number of expert raters, variability in scoring patterns, and the heterogeneity of evaluator perspectives. While the overall trends and high mean PEMAT and GQS scores support the main conclusions of the study, the breadth of these intervals suggests caution when generalizing the findings. To address this

limitation, future research will involve a larger and more diverse panel of experts, employ stratified sampling to balance evaluator characteristics, and perform additional reliability testing (e.g., test–retest) to achieve narrower and more stable confidence intervals.

As this study focused on assessing the baseline feasibility and quality of AI-generated materials, no structural revisions were implemented. However, the findings highlight the need for future enhancements—such as incorporating clearer visuals, patient checklists, and scenario-based instructions—to improve the material's actionability and achieve broader expert consensus.

The intraclass correlation coefficient (ICC) of 0.84 indicates strong agreement among expert evaluations. However, the relatively wide confidence interval (95% CI: 0.271–0.927) suggests some variability in how individual experts assessed the material. This may reflect differences in the interpretation of visual or instructional elements, underscoring the need for clearer design or evaluator calibration.

Future studies should consider involving a larger number of experts to reduce variability and enhance inter-rater reliability. Similar patterns have been observed in other evaluations of ChatGPT-generated educational content, where accuracy, completeness, and readability varied across reviewers [46–48]. Notably, content produced with ChatGPT-4.0 tends to be more usable and clearer when guided by well-structured, specific prompts.

A recent study on the use of ChatGPT in nursing practices reported that chatbots were perceived as more convenient and user-friendly than traditional applications. It was also noted that nurses' familiarity with other technologies contributed to higher adoption rates. These findings suggest that AI tools can be effectively integrated into nursing workflows, provided that appropriate support and training are available [49].

The effectiveness of AI-generated educational materials—particularly their readability, comprehensibility, and actionability—largely depends on the specificity and quality of the prompts used. As AI-generated content becomes increasingly prevalent in healthcare, it is essential that such materials are reviewed and validated by healthcare professionals to ensure their accuracy and usability. Further research is needed to optimize prompt strategies, ensure content reliability, and explore the practical integration of AI tools in patient education.

In this study, the Turkish readability score (Ateşman Index) indicated that the material was easy to understand, whereas the Flesch Reading Ease (FRE) score—developed for English texts—classified it as very difficult. This discrepancy likely stems from the agglutinative structure of the Turkish language, which increases average word and syllable length compared to English [27,

28]. As such, formulas originally designed for English may not accurately reflect the readability of Turkish texts.

Local measures like the Ateşman Index offer a more realistic assessment for Turkish-language materials. However, reporting both local and international metrics is essential for cross-cultural comparisons and broader dissemination of AI-generated content. Addressing this contradiction may involve refining AI prompts and making linguistic adjustments during the content creation process.

Future studies should explore multilingual and culturally adapted readability strategies when developing AI-based educational materials. Ultimately, while AI holds promise in creating effective patient education content, its success depends on the integration of linguistic, cultural, and professional oversight to ensure accuracy, clarity, and patient relevance.

Strengths of the study

A notable strength of this study is the development of an AI-supported educational material tailored to Turkish-language discharge education following total hip replacement, addressing a gap in existing AI-based health education studies by focusing on a non-English, specialty-specific context. The comprehensibility, applicability, and quality of the developed material were evaluated using valid and reliable tools such as the Patient Education Materials Assessment Tool (PEMAT) and the Global Quality Scale (GQS). Additionally, high internal consistency (Cronbach's $\alpha=0.76$) among expert reviewers supports the reliability of the evaluation process.

The increasing use of AI tools such as ChatGPT in healthcare education underscores the need to address potential risks, including hallucinations, factual errors, and safety concerns. In our study, we mitigated these risks through a rigorous content verification process involving expert review and cross-referencing with high-quality, up-to-date clinical evidence. This multi-layered approach ensured that the final educational materials were accurate, reliable, and safe for patient use. Nevertheless, we acknowledge that AI outputs may still contain subtle inaccuracies, and we recommend that future studies incorporate ongoing monitoring and updates to maintain accuracy as clinical guidelines evolve.

Limitations

Despite these strengths, the study has several limitations. First, the AI-generated material is considerably lengthy, which may be burdensome for expert evaluation. Second, the number of visuals produced by the chatbot is limited. Most importantly, a key limitation is the lack of patient involvement in the validation process. Although experts rated the material highly for understandability and actionability, no direct feedback was obtained

from patients—the ultimate end-users of the material. Without incorporating patient perspectives, especially those with varying levels of health literacy, the findings may not fully reflect real-world usability or effectiveness. Therefore, the reliability of the materials in actual clinical settings remains uncertain. Additionally, the readability of the educational content was assessed using a single online readability calculator. As noted in recent literature, such as the study by Ozduran et al. (2025), using only one tool can produce inaccurate or inconsistent results due to algorithmic differences across platforms. Ozduran and colleagues recommend using more than one readability calculator and averaging the results to increase reliability [50]. Future studies should consider adopting this approach to obtain a more accurate readability evaluation.

Lastly, because this study was conducted with a limited number of experts and without direct patient testing, the generalizability of its findings to broader patient populations and different clinical contexts is limited.

To address this gap, a pilot study involving a representative patient cohort is planned as a follow-up. This future study will assess real-world comprehensibility and usability of the educational material and will be conducted with institutional ethical approval and informed consent.

Conclusion

This study found that ChatGPT provides an accurate, understandable, applicable, and high-quality total hip replacement discharge education material in terms of content, but the readability level of this material is relatively difficult. AI-supported applications have the potential to revolutionize post-surgical patient education and patient interaction processes.

Additionally, to address the issue of end-user validation, we plan to conduct a pilot test with a patient cohort in a follow-up study. This real-world evaluation will provide critical feedback on the comprehensibility and actionability of the educational material. Ethical approval and informed consent will be obtained prior to implementation. This step aims to ensure that the material is not only expert-approved but also patient-centered and practically applicable.

The results obtained are considered an important step to facilitate and encourage the preparation of patient education materials by clinical and academic nurses. Patient education, which has a critical role in preventing post-surgical complications, and the methods developed to provide this education have gained a new dimension with the advancement of technology and AI. This research aims to support the integration of advanced technologies such as AI into patient education practices and contribute to maximizing the benefit of these technologies.

Abbreviations

AI	Artificial Intelligence
OECD	Organisation for Economic Co-operation and Development
THR	Total Hip Replacement
SMOG	Simple Measure of Gobbledygook
ARI	The Automated Readability Index
FKGL	Flesch-Kincaid Grade Level
FRE	Flesch Reading Ease
ARLC	Average Reading Level Consensus
PEMAT	Patient Education Materials Assessment Tool
PEMAT-A/V	Audiovisual Materials Patient Education Materials Evaluation Tool
PEMAT-P	Printable materials patient education materials evaluation tool
QOS	Quality score of the education materials
SPSS	Statistical Package for Social Science

Acknowledgements

We would like to thank the Proofreading & Editing Office of the Dean for Research at Erciyes University for the copyediting and proofreading service for this manuscript.

Authors' contributions

HT contributed to conception, design, critically revised the article, writing, editing of manuscript and final approval of manuscript. GMB contributed to conducted the interview, data analysis, final approval of manuscript. YS contributed to conducted the interview, data analysis, final approval of manuscript.

Funding

The authors received no financial support for the research, authorship, and/or publication of this article.

Data availability

No datasets were generated or analysed during the current study.

Declarations

Ethics approval and consent to participate

This study did not require ethics committee approval because it used the open-access ChatGPT application. Experts were also informed about the study's content and agreed to participate.

Consent for publication

This study does not contain data from any individual person.

Competing interests

The authors declare no competing interests.

Received: 25 February 2025 / Accepted: 19 October 2025

Published online: 24 November 2025

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